ClaimLab Attention Lab - Section A: From scores to weights

Softmax turns a finite vector of scores into nonnegative weights summing to one.

Increasing all scores by the same amount preserves the weights.

Multiplying unequal scores changes their relative influence.

Equal scores receive equal weights.

A numerical investigation can compare score vectors before and after multiplication. Its inputs and results should be visible so the learner can explain the relationship rather than memorize a picture.

ClaimLab Attention Lab - Section B: Attention ingredients

In a small attention example, a query is compared with keys to produce scores.

A normalization step produces weights. Those weights combine values to produce an output.

The objects participating in scoring and those contributing output information have different roles.

A synthetic dot-product experiment can investigate the effect of score scaling under stated assumptions. Synthetic vectors do not represent a trained model’s learned behavior or reproduce a translation benchmark.

ClaimLab Attention Lab - Section C: Which positions may contribute?

In a causal attention illustration, a position can use its current and earlier positions while future positions are excluded.

A mask changes which entries contribute before row-wise normalization.

Each row in this example retains at least one allowed entry.

This explains one operation. A full language-model training procedure includes additional design choices and should not be inferred from the illustration alone.